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Play a Part in Parkinson's Research

The role of LRRK2 kinase activity in human macrophage function

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Summary

LRRK2 is highly expressed in immune cells such as macrophages but the precise function is unknown. Parkinson's disease is commonly associated to pathogenic mutations and the most common one is the LRRK2 G2019S. Understanding how these mutations affect the immune function of macrophages is important to understand the immune component in Parkinson's disease. We obtained stem cells from patients harboring the LRRK2 G2019S mutation and by using genetic engineering, we restored the mutation. Then, stem cells were differentiated into macrophages to study the impact of the mutation in immune cells. These cells were subsequently stimulated with several immunomodulators and infected with bacterial pathogens to determine if cells responded differently.

Method

Obtention of iPSCs

We obtained two lines from the PPMI resource that were grown and banked following protocols established in the lab after testing negative for mycoplasma. Both lines grew well and were prepared for quality check. The selected clones were 11306.104 (46XY) and 11348.102 (46XX) and the LRRK2 gene analysed by Sanger sequencing.

Gene editing to generate isogenic iPSC

Isogenic gene-corrected controls were obtained using the CRISPRs/Cas9 system. We designed a 2 steps approach where in a first step: select clones with InDel mutation only in the "mut allele" (G2019S) and in a second step Design a new sgRNA to target only the targeted allele in the selected clone. We then selected clones for the "wt/wt" allele event. As a result, we obtained two sets: the 11306.104 (LRRK2-G2019S) and LRRK2-E10 (Isogenic control) and 11348.102 (LRRK2-G2019S) and LRRK2-G10 (Isogenic control). Mycoplasma, Sanger sequencing and Karyostat quality checks were performed in both clones. Additionally, LRRK2-E10 clone was tested for pluritest and low pass sequencing. All clones were genotyped and banked for long-term storage.

Differentiation into iPSC-derived macrophages (iPSDM) and characterization

The 11306.104 (LRRK2-G2019S) and LRRK2-E10 (Isogenic control) and 11348.102 (LRRK2-G2019S) and LRRK2-G10 (Isogenic control) were used to generate iPSDM following a previously reported protocol using embryonic bodies (EB)(van Wilgenburg et al., 2013) (Bernard et al., 2020). Briefly, the forming EB were used to produce monocytes in the presence of hM-CSF and hIL-3 and cells are grown in serum free and feeder free media.

Flow cytometry

1 x 10^6 iPSDM and 1 x 10^6 monocytes differentiated from LRRK2-G2019S and LRRK2-E10 clones were pelleted by centrifugation (350 x g for 5 minutes) and resuspended in PBS with fluorochrome-conjugated antibodies for 30 minutes at room temperature. The following antibodies were used: Alexa Fluor® 488 Mouse Anti-Human CD14 (562689, BD Biosciences), Alexa Fluor® 647 Mouse Anti-Human CD16 (557710, BD Biosciences), PE Mouse Anti-Human CD119 (558934, BD Biosciences), FITC Mouse Anti-Human CD163 (563697, BD Biosciences), APC Mouse Anti-Human CD206 (561763, BD Biosciences), PE Mouse Anti-Human Siglec-1 (CD169) (565248, BD Biosciences) and BV421 Mouse Anti-Human CD86 (562433, BD Biosciences). Isotype control antibodies used were: Mouse IgG1κ-PE (555749, BD Biosciences), Mouse IgG2bκ-Alexa647 (557903, BD Biosciences), Mouse IgG1κ – FITC (554679, BD Biosciences), Mouse IgG1κ-BV421 (562438, BD Biosciences) and Mouse IgG1κ – APC (555751, BD Biosciences). Cells were then washed in PBS and resuspended in 500uL PBS for acquisition. Data were acquired using an BD LSR Fortessa™ Cell Analyzer (BD Biosciences) and analysed using FlowJo software (FlowJo, LLC, v10.8.0). At least 10,000 events per condition were recorded.

Cell treatments and cytokine measurement

6 x 10^5 iPSDM were seeded in 96 well olefin-bottomed plates. Cells were treated with for 24 hours with LPS 250 ng/mL (NBP2-25295, Bio-techne) or PAM3CSK4 10ng/mL (4633/1, Bio-techne) in the presence or absence of MLi-2 100nM (5756/10, Tocris). 100uL supernatant was collected and stored at -80°C. Cytokine levels were then analysed using the magnetic Bio-Plex Pro Human Cytokine 27-plex kit (M500KCAF0Y, Bio-rad) as per the manufacturers' instructions. Data shown is the mean cytokine level in pg/mL from three independent experiments. Cytokines outside the limit of detection are indicated as OOR > or OOR <.

PluriTest and KaryoStat Assay

2 x 10^6 iPSCs were centrifuged at 300g for 3 minutes. The resulting pellet was then stored at -20°C until ready to be sent to the Thermo Fisher company for the PluriTest and KaryoStat Assay.

Sanger Sequencing

1 x 10^6 iPSCs and iPSDM were centrifuged at 300g for 3 minutes. The resulting pellet was subjected to genomic DNA (gDNA) extraction using the DNeasy Blood & Tissue kit (#69504 Qiagen).

PCR was then performed on the extracted gDNA using the specified primers: forward primer (5' – AAAATTGGGTCTTTGCCTGAGATAG-3') and reverse primer (5'-TCACAAGTGCCAACAATACCTAGAC-3'). After the PCR amplification, the PCR product was purified using the QIAquick PCR purification kit (#28104 Qiagen).

Subsequently, the purified PCR product was sent for Sanger sequencing analysis using the following sequencing primers: 5'- TATATTTAAGGAAATTAGGAC-3' and 5'-TTCACAATGTGATAGACTCTG-3'.

Mycoplasma test

1 x 10^6 iPSCs and iPSDM were centrifuged at 300g for 3 minutes. The spent media containing non-adherent cells was removed, and the cell pellets were collected in 1.5 ml Microcentrifuge Eppendorf tubes. The supernatant was discarded, and the cell pellets were stored at -80°C until they were ready to be sent for Mycoplasma PCR-based testing using the Universal Mycoplasma Detection Kit (Catalog Number 30-1012K).

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Low-pass Whole Genome Sequencing

DNA samples, with a concentration of 5 ng/µl, were subjected to Low-pass Whole Genome Sequencing. The sequencing was performed using approximately 16 million reads for each sample, resulting in around 0.5-1x coverage. This coverage level is adequate for robust karyotyping analysis. The samples were prepared using the Nextera Flex protocol for the sequencing process.

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